

[REDACTED]

From: Steve Greer [REDACTED]
Sent: Saturday, February 25, 2012 12:13 PM
To: Ted Loder
Cc: [REDACTED]
Subject: Re: Analysis of [REDACTED]

Thanks [REDACTED]

[REDACTED] file..steve

On Fri, Feb 24, 2012 at 3:15 PM, Ted Loder [REDACTED] wrote:
Hi All,

Some further information re [REDACTED] technology. I just spoke with him again and he answered all of my questions at this point.

In yesterday's analysis, I hypothesized that the greater efficiency he was observing was due to the comparison of an inefficient internal combustion engine (at 30-35%) vs. essentially 100% efficiency of burning pure H₂.

However, he used a \$3000 gas flow meter specifically designed for H₂ gas and actually measured the gas output as did a professor from Vanderbilt University and found that the rate of gas production was about 400 cu ft. of H₂ per 6-7 lbs of Al as he mentioned the other day.

He also did a comparison test using a 60 HP Ford engine. It ran at 1200 rpm for 4 hours and 20 minutes using 5.7 lbs of Al. He then ran it using 2 commercial bottles of H₂ gas (each containing about 200 cu ft. of gas) which was 400 cu ft of H₂ gas which is equivalent to about one gallon of gas. It ran for 4 hours on the 2 tanks of the gaseous H₂.

Conclusions based on the comparison test. It appears that about 5.7 lbs of Al produced a little more of H₂ gas than was contained in 400 cu ft. equivalent to about one gallon of gas. This is significantly less than the 20 lbs of Al required to produce the same amount based textbook chemistry.

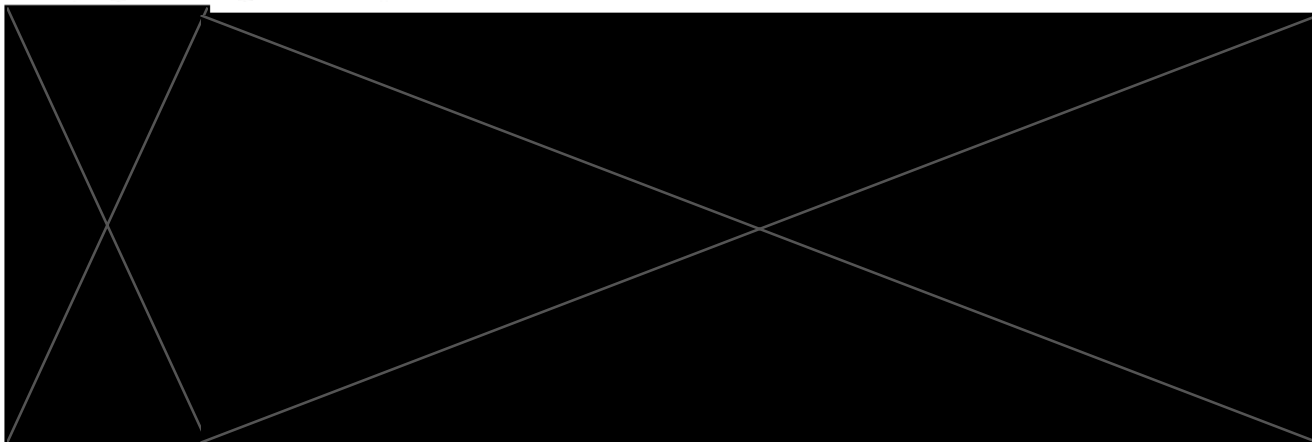
So whatever he is doing produces nearly 3.5 times more H₂ gas than standard textbook chemistry ($20/5.7 = 3.5x$). The vessel he is using is made of carbon steel and coats up with the alumina as the reaction proceeds (something he has not noted to others at this point). In other words standard chemistry does not explain what he is observing (there must be some other chemical reactions occurring that we are not accounting for). If we couple this increased efficiency with the increased burning efficiency, the whole process becomes very efficient, meaning that it would be a viable technology to continue to assess.

That's it for now.

[REDACTED]

On Feb 23, 2012, at 2:41 PM, Steve Greer wrote:

Ultimately, we will need to go to his place and assiduously do ALL of the empirical measurements ourselves--relying none on pre-existing representations and measurements. Steve



that is



Notes Regarding



Al/H₂ Technology

Feb. 23, 2012

See background information at the end of the discussion for the chemical reactions I discuss below. 

So 2 moles of Al produces 3 moles of H₂ gas. MW of Al = 26.982 g/mole.

3 moles of H₂ gas = 22.4 liters x 3 = 67.2 liters at STP

then 1 mole of Al metal = 26.982 g/mole produces 1.5 moles of H₂ gas or 1.5 x 22.4 = 33.6 liters

There are 453.592 g/lb / 26.982 g/mole Al = 16.81 moles of Al per lb. Since each mole of Al produces 1.5 moles of H₂ gas, then

16.81 x 33.6 liters = 564.82 liters or 19.95 cu ft. at STP produced by 1 lb. of Al

Since 400 cu ft is equivalent to 1 gal of gasoline according to , then it should take about 20 lbs of Al to produce the equivalent of 1 gallon of gas. Now he states that it takes 6.1 to 7 lbs to produce 400 cu ft of H₂ gas, whereas it calculates to 20 lbs. He did tell me that experts told him it should take 19-21 lbs. per gallon of gasoline which agrees with my analysis.

Something does not make sense.

Although he does not state it, the H₂ gas may be "burning" at 100% efficiency while normal gasoline only burns at 30-40 % efficiency. Which could account for the difference. In other words, the amount of H₂ gas produced from 7 lbs of Al, burned at 100% efficiency would be equivalent to 1 gallon of gas burned at 30-40 % efficiency.

7 lbs produces 7 x 19.95 cu ft = 139.65 cu ft. which is about 35% of the 400 cu ft equivalent to a gallon of gas.

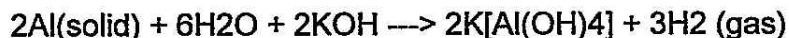
If my analysis is correct then it appears that his system obeys the chemistry equations as we understand them, but the H₂ gas burns at 100 % efficiency, which is where he gets his apparent advantage.

Where do we go from here??

Background information:

From: <http://www.newton.dep.anl.gov/askasci/chem03/chem03682.htm>

The equation that you are looking for is:



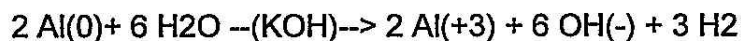
This is typically an intermediate step in the production of alum by further addition of sulfuric acid.

Hope this helps.

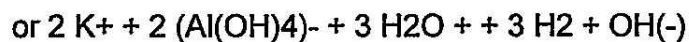
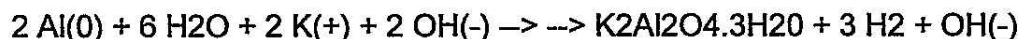


This is a famous reaction. Aluminum is so reactive, so energetic, it is the most common metal added to rocket fuel. It would always react with water, if it did not tend to grow an oxide film that blocks off the surface and stops the reaction. Aluminum oxides are soluble in basic water solutions, so KOH enables the reaction to go continuously.

Basically, it is:



There are many slightly different ways to express it which more nearly describe the exact state of the stuff in water after the reaction. This is where the KOH gets involved a little, and where one might understand why precipitates form.



or a mix of these and similar things.

This reaction consumes some OH⁻ as well as Al and water. I would not assume that K₂Al₂O₄·3H₂O is exactly what is there, but at least the equation is balanced.

The hydrogen evolved is flammable and explosive. Please only do this with supervision and use no glass containers. Use ventilation. The alkaline water-mist is bad for lungs eyes and skin, too.



Information about [REDACTED] Al/Hydrogen Technology)
Review of issues by [REDACTED] for TOP

Raw Materials

Aluminum. In theory any Al can be used. [REDACTED] uses Al pipe (2.5" OD, 2" ID) since it can be cut to length and easily moved in and out of the KOH solution to control rate of H₂ production. Solution level can also be adjusted to control rate. Great for demonstration purposes.

KOH (25%). Used to adjust the pH to 13-14. Cost is minimal and it is reused according to [REDACTED]

Conversion Numbers:

[REDACTED] is using both a 4 and 10 KW generator for testing as well as other systems. We focus here on a nominal 10KW system.

6.1 to 7 pounds of Al is needed to produce 400 cubic feet of H₂ gas equivalent to one gallon of gasoline or about 125,000 BTUs.

Estimating fuel needs: A Winco 10.8 kw generator uses 15gal per 9 hour full load or 1.7gal/hour to produce approx. 10KWH. Another generator (Generac) uses 1 gal/hour to produce 5 KW. This estimates that it takes approx. 0.17 to 0.2 gal to produce one KWH. Other generators may be more efficient. Let us assume 0.20 gal per hour per KWH. This is equivalent to 7 lbs x 0.2 = 1.4 lbs of Al to produce one KWH.

Relative Costs:

Assume gasoline is \$4.00/gallon (approx. retail price in Jamaica) and Al is \$0.15 to 0.40/lb. Then at 7 lbs Al per gallon of gasoline produced it costs between \$1.05 and \$2.80 of Al to produce one gallon of gas or about 5 KWH using small generators. Larger generators may be more efficient.

Product Value:

Approx. 2.7 lbs of alumina are produced for every lb. of Al used. The alumina produced by this technology is a very fine uniformly sized product in the nano or sub-micron size range. This alumina is retail valued at

between \$1.45 to over \$2000 per lb depending on purity and particle size. So even selling just a portion of the alumina produced would significantly reduce the cost of producing the KWH. Thus the alumina can be sold commercially or recycled back to pure Al for an energy cost of approx. 1/3 KWH per pound of Al.

Hydrogen gas (high purity H₂). The method can produce hydrogen gas up to a pressure of 3000psi. Presently hydrogen gas sells for est. \$65/tank equivalent to approx. 200 cu ft or 1/2 gallon of gasoline. This could be a very lucrative side business.

Factors determining the cost effectiveness of this method.

1. Present commercial costs of electrical power at the chosen location.
2. Cost of conversion of present diesel generators to use either 100% hydrogen or 90% hydrogen and 10% diesel (not discussed here).
3. Availability of low cost scrap Al in easily used form. It may be necessary to form the Al into shape to be easily used in the reactor (pipe, pellets, powder?). Although recyclable, there are costs associated with it so these costs are critical.
4. Market for the alumina and/or alumina to Al recycling facilities near by and cost effective. How the alumina is used or sold will be one of the deciding factors in the overall cost savings analysis.
5. Value of excess hydrogen gas on local market.
6. Other?

Demonstration Project must include:

1. Site specific expenses and cost issues and local energy costs.
2. Method of recovering some production cost through sale of alumina and excess hydrogen gas to local markets.
3. Building a larger gas generator. At present ~~existing~~ system produces 30 cu ft. per minute or 1800 cu ft per hour. This is equivalent to approx. 4.5 gal. of gasoline (worth \$18) per hour producing approx. $4.5/0.2 = 22.5$ KWH using approx. $7 \times 4.5 = 31.5$ lbs of Al (est. cost of \$0.15 - 0.40) or \$5.00 - \$12.60.
4. Determination of the optimal size of the alumina and markets for commercial sale at highest price.